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F.E. EXAMINATION, 2019
ENGINEERING MATHEMATICS—II
(2015 PATTERN)

Time : Two Hours

Maximum Marks : 50

- N.B. :—**
- (i) Attempt Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4, Q. No. 5 or Q. No. 6, Q. No. 7 or Q. No. 8.
 - (ii) Neat diagrams must be drawn wherever necessary.
 - (iii) Figures to the right indicate full marks.
 - (iv) Use of electronic pocket calculator is allowed.
 - (v) Assume suitable data, if necessary.

1. (a) Solve the following differential equations :

(i) $(1 + \log xy)dx + \left(1 + \frac{x}{y}\right)dy = 0$ [4]

(ii) $x^4 \cdot \frac{dy}{dx} + x^3 y - \sec(xy) = 0$ [4]

- (b) A long hollow pipe has an inner diameter of 10 cm and outer diameter of 20 cm. The inner surface is kept at 200°C and the outer surface at 50°C. The thermal conductivity is 0.12. Find the temperature at a distance 7.5 cm from the centre of the pipe. [4]

Or

2. (a) Solve $3y^2 \frac{dy}{dx} + 2xy^3 = 4xe^{-x^2}$ [4]

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- (b) (i) A body at temperature 100°C is placed in a room whose temperature is 20°C and cools to 60°C in 5 minutes. Find its temperature after a further interval of 3 minutes. [4]
- (ii) A resistance of 100 ohms, an inductance of 0.5 henry are connected in a series with a battery of 20 volts. Find the current in the circuit if $i = 0$ at $t = 0$. [4]
3. (a) Obtain the Fourier series expansion of $f(x) = x^3$, $-\pi < x < \pi$. [5]
- (b) Evaluate $\int_0^{\infty} \sqrt{y} e^{-\sqrt{y}} dy$. [3]
- (c) Trace the following curve (any one): [4]
- (i) $y^2(2a - x) = x^3$.
- (ii) $r = a \cos 3\theta$
- Or
4. (a) If $I_n = \int_{\pi/4}^{\pi/2} \cot^n \theta d\theta$, prove that : [4]
- $$I_n = \frac{1}{n-1} - I_{n-2}$$
- (b) Prove that $\int_0^{\infty} \frac{e^{-ax} - e^{-bx}}{x} dx = \log\left(\frac{b}{a}\right)$ [4]
- (c) Find the length of upper arc of one loop of lamniscate $r^2 = a^2 \cos 2\theta$ [4]
5. (a) Find the centre and radius of the circle of intersection of the sphere $x^2 + y^2 + z^2 - 2y - 4z - 11 = 0$ by the plane $x + 2y + 2z = 15$. [5]

(b) Find the equation of right circular cone which has its vertex at the point $(0, 0, 10)$ and whose intersection with the plane XOY is a circle of diameter 10. [4]

(c) Find the equation of right circular cylinder whose axis is $x = 2y = -z$ and radius is 4. [4]

Or

6. (a) Find the equation of sphere through the circle $x^2 + y^2 + z^2 = 1$, $2x + 3y + 4z = 5$ and which intersects the sphere $x^2 + y^2 + z^2 + 3(x - y + z) - 56 = 0$ orthogonally. [5]

(b) Find the equation of right circular cone which passes through the point $(2, -2, 1)$ with vertex at the $(0, 0, 0)$ and axis parallel to the line $\frac{x-2}{5} = \frac{y-1}{1} = \frac{z+2}{1}$. [4]

(c) Find the equation of right circular cylinder of radius 2 whose axis passes through $(1, 2, 3)$ and has direction cosines proportional to $2, -3, 6$. [4]

7. Attempt any *two* of the following :

(a) Evaluate by changing the order of integration :

$$\int_0^1 \int_0^{\sqrt{1-y^2}} \frac{\cos^{-1} x \, dx \, dy}{\sqrt{(1-x^2-y^2)(1-x^2)}} \quad [7]$$

(b) Find the volume cut-off from the paraboloid $x^2 + \frac{y^2}{4} + z = 1$ by the plane $z = 0$. [6]

(c) Find the moment of inertia of one loop of Lemniscate $r^2 = a^2 \cos 2\theta$ about initial line. [6]

Or

8. Attempt any *two* of the following :

(a) Find the area inside the circle $r = a \sin \theta$ and outside the cardioid $r = a (1 - \cos \theta)$. [7]

(b) Evaluate :

$$\iiint_V \frac{dx dy dz}{\sqrt{1-x^2-y^2-z^2}} \quad [6]$$

taken throughout the volume of the sphere $x^2 + y^2 + z^2 = 1$ in the positive octant.

(c) Find the centre of gravity of an area of the cardioid $r = a (1 + \cos \theta)$ [6]